The Augmented Synthetic Historical Control Method

Introduction

Causal inference in economic and policy research often relies on comparing a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics, second only to difference-in-differences. The idea of SC is straightforward: the untreated potential outcomes of a treated unit are approximated by a weighted average of donor units. Difference-in-differences relies on the assumption of parallel pre-intervention trends [Baker et al., 2025].

In many settings, however, no suitable donor units are available—for example, during global events such as pandemics or financial crises, or when no valid donor units exist. When no valid untreated donor units are available, researchers often turn to the interrupted time series (ITS) design. In its canonical form, ITS models the pre-treatment outcome path as a parametric linear trend and tests for changes in level or slope following an intervention. In fact, even comparative interrupted time series designs have been used by policy analysts [Coopersmith et al., 2022]. Even so, ITS can be fragile in many realistic applications. Its validity depends critically on the assumption that an OLS line fitted to the pre-treatment period provides a reasonable counterfactual projection. In practice, many outcome series feature nonlinear dynamics, seasonality, or structural breaks that violate this assumption. Moreover, ITS typically imposes a homogeneous effect structure, assuming a single jump in level and/or slope that applies equally across all post-treatment periods. These restrictions can lead to biased inference when the true effects are time-varying or when pre-treatment is poor.

To address this shortcoming, Chen et al. [2024] proposed the synthetic historical control method (SHC). Rather than relying on external donors, SHC constructs a donor pool from the treated unit's own pre-treatment history, splitting the pre-treatment period into overlapping segments. The SHC also, unlike ITS, uses kernel smoothing to account for non-linearities in the pre-treatment time series data which ITS does not naturally account for. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, allowing for effect size heterogeneity found in SCM and increasingly DID but not ITS. However, like Abadie et al. [2010, 2015], SHC restricts weights to the simplex. This guarantees interpretability but can be too restrictive when noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the Augmented Synthetic Historical Control method (ASHC), which extends SHC in two key ways. First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, allowing controlled extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, ASHC adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM [Ben-Michael et al., 2021]. This encourages the estimator to remain close to SHC unless extrapolation is needed. ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where exact pre-treatment fit is otherwise impossible.

The paper makes three contributions. Methodologically, it introduces the ASHC estimator, which generalizes SHC by permitting limited, penalized extrapolation while maintaining the stabilizing Gram penalty. Theoretically, it derives finite-sample error bounds that decompose ASHC prediction error into four interpretable parts: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. It further establishes asymptotic consistency and a convergence rate of $O_p(m^{-H/(2H+3)})$, which approaches the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order H increases. Practically, this means that when the pre-treatment length m is large (e.g. $m = 120$) and H grows, the convergence rate becomes practically close to the parametric $O_p(m^{-1/2})$, so ASHC achieves efficient performance in realistic sample sizes.

While the finite-sample analysis provides guarantees on the accuracy of ASHC for realistic data sizes, asymptotic results are equally important. The finite-sample bounds show how ASHC predictions may differ from SHC depending on regularization strength, operator norm, smoothness bias, and noise. The asymptotic results complement this by demonstrating that, as the amount of pre-treatment data grows, ASHC's prediction error vanishes at a near-parametric rate under mild conditions. Together, these perspectives ensure that ASHC is both practically reliable in finite samples and theoretically sound in large-sample limits.

The discussion begins with a careful review of SHC, its assumptions, and its implementation, before introducing the ASHC framework. We then turn to the theoretical analysis, establishing finite-sample error bounds and asymptotic properties. The empirical performance of ASHC is explored through calibrated simulations, followed by an application that illustrates its usefulness in practice. Proofs are collected in the appendix, and all computations are carried out using Python's mlsynth library (https://mlsynth.readthedocs.io/en/latest/).

For policy analysts, the message is straightforward: ASHC provides a principled way to construct reliable counterfactuals even in environments where no external donor units are available. Furthermore, it allows policy analysts to do this even when methods such as interrupted time series method or the Chen et al. [2024] would not do well.

The remainder of the paper proceeds as follows. Section situates our contribution within the existing literature on causal inference in policy evaluation. We then review the SHC method in Section before introducing the augmented SHC (ASHC) in Section . Section presents our theoretical results, including finite-sample error bounds and asymptotic properties. We then turn to an empirical application in Section , replicating the analysis of Barcelona's 2015 hotel moratorium from Bel et al. [2021], a setting that allows us to compare ITS, SCM, SHC, and ASHC on equal footing. Finally, Section concludes with implications for both methodological research and policy analysis.

Literature Review

Methodology

The Synthetic Historical Control Method

The Augmented Synthetic Historical Control Method

Theoretical Findings

Empirical Application

We replicate the setting of Bel et al. [2021], who study the effects of Barcelona's 2015 hotel moratorium using data on hotel prices from 2010 to 2017. This application provides a compelling case to benchmark ITS, SCM, SHC, and ASHC against one another.

Conclusion

References

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505, 2010. URL https://doi.org/10.1198/jasa.2009.ap08746.

Alberto Abadie, Alexis Diamond, and Jens Hainmueller. Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2):495–510, 2015. doi: https://doi.org/10.1111/ajps.12116.

Andrew Baker, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. Difference-in-differences designs: A practitioner's guide, 2025. URL https://arxiv.org/abs/2503.13323.

Germà Bel, Stephan Joseph, and Ferran A. Mazaira-Font. The effects of moratoriums on hotel building: An anti-tourism measure, or rather protection for local incumbents? *Applied Economics Letters*, 29(14): 1319–1324, 2021. doi: 10.1080/13504851.2021.1927958.

Eli Ben-Michael, Avi Feller, and Jesse Rothstein. The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536):1789–1803, dec 2021. doi: 10.1080/01621459.2021.1929245.

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.

Jared Coopersmith, Thomas D. Cook, Jelena Zurovac, Duncan Chaplin, and Lauren V. Forrow. Internal and external validity of the comparative interrupted time-series design: A meta-analysis. *Journal of Policy Analysis and Management*, 41(1):252–277, 2022. doi: https://doi.org/10.1002/pam.22361. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/pam.22361.